Saturn (RevA)

Date: 3/12/15

Introduction:

The Saturn board is a stripped down version of the MAD_Oreo door sensor board. This unit strips away the multiple functions of Oreo to provide a smaller footprint, capable of transmitting via ZigBee, whether or not a door/window is closed. This board is based on the EM3585 micro-controller, implemented as a discrete design, allowing direct interfacing with the antenna. Other components used on the board include a Hall Effect Sensor (MRMS511H), a thermistor (NTCLE203), a tamper switch (SPVM110200), a reset switch, and a tri-color LED (LTST-C19HE1WT).

Hardware Description:

The Jupiter board is based on the EM3585 micro-controller operating at 24 MHz. The CPU interfaces directly to 1 or 2 switches, 1 tri-color LED, 1 Hall Effect Sensor, and 1 Thermistor. A JTAG port is also available for programming/debugging the internal CPU Flash memory. The board is powered by a 2450 Coin Cell Battery, supplying +3V. This device uses a ceramic balun in its TX/RX circuit.

The pin assignments for the JTAG Port (J1):

Pin 1: VDD (3V)
Pin 2: JTDO
Pin 3: JRST
Pin 4: JTDI
Pin 5: GND
Pin 6: JTCK
Pin 7: JTMS
Pin 8: RESET
Pin 9: PTI_EN
Pin 10: PTI_DATA

Pin listings for GPIO lines on the EM3585:

PA0: NC
PA1: NC
PA2: NC
PA3: NC
PA4: PTI_EN
PA5: PTI_DATA
PA6: NC
PA7: GREEN_LED (“1” to turn LED off, “0” to turn LED on)
PB0:THERM_VDC (Voltage output for thermistor divider circuit)
PB1: NC

PB2: NC
PB3: NC
PB4: READ_TAMPER
PB5: Reset Switch
PB6: RED_LED (“1” to turn LED off, “0” to turn LED on)
PB7: BLUE_LED (“1” to turn LED off, “0” to turn LED on)
PC0: JRST
PC1: THERM_READ (Value from thermistor divider circuit)
PC2: JTDO
PC3: JTDI
PC4: JTMS
PC5: NC
PC6: DOOR_OUT
PC7: NC